

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	35.8	12	1.7	0.7	0.3	49.5
U3 - Toxicity prior Beta(0.15, 0.85)	35.9	11.7	1.5	0	0.1	50.8
U2 - Toxicity prior Beta(0.3, 0.7)	36	14.2	3	0.7	0.4	45.7
U3 - Toxicity prior Beta(0.3, 0.7)	38	13.3	1.9	0.2	0	46.6
U2 - Toxicity prior Beta(0.5, 0.5)	36.3	17.2	3	0.6	0.5	42.4
U3 - Toxicity prior Beta(0.5, 0.5)	37.8	16.1	2	0.1	0.2	43.8
U2 - Toxicity prior Beta(1, 1)	37.7	15.9	3.3	0.6	0.3	42.2
U3 - Toxicity prior Beta(1, 1)	39.1	15.2	1.9	0.1	0	43.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	14.6	6.2	2.3	1.3	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	14.8	6	2.2	1.3	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	14.2	6.4	2.7	1.5	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	14.6	6.1	2.5	1.5	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	13.6	6.6	3.1	1.7	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	14.1	6.4	2.9	1.6	0.6	
U2 - Toxicity prior Beta(1, 1)	13.7	6.6	3	1.7	0.6	
U3 - Toxicity prior Beta(1, 1)	14.2	6.2	2.9	1.7	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.3	3.2	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	1.2	3	0.6	0.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	1.2	3	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	1.2	2.9	0.6	0.4	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	1	2.7	0.5	0.5	0	
U3 - Toxicity prior Beta(0.5, 0.5)	1	2.7	0.5	0.4	0.1	
U2 - Toxicity prior Beta(1, 1)	1	2.1	0.5	0.4	0	
U3 - Toxicity prior Beta(1, 1)	1.1	2	0.5	0.4	0	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	27.6	23.4	35.6	11.7	1.1	0.6
U3 - Toxicity prior Beta(0.15, 0.85)	32.4	22.2	34.4	9.5	0.9	0.6
U2 - Toxicity prior Beta(0.3, 0.7)	25.5	21.6	36.2	13.8	2.1	0.8
U3 - Toxicity prior Beta(0.3, 0.7)	29.9	21.5	35.1	11.3	1.4	0.8
U2 - Toxicity prior Beta(0.5, 0.5)	23.1	20.3	39.2	14.3	2.5	0.6
U3 - Toxicity prior Beta(0.5, 0.5)	27.7	22.3	34.8	12.9	1.6	0.7
U2 - Toxicity prior Beta(1, 1)	23.1	19.2	38.6	15.4	3.1	0.6
U3 - Toxicity prior Beta(1, 1)	27.9	19.7	35.5	14.3	1.8	0.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.4	8	8.4	3.4	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	10	8.1	8.3	3.1	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	8.8	7.8	8.9	3.6	0.9	
U3 - Toxicity prior Beta(0.3, 0.7)	9.3	7.9	8.6	3.5	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	8.1	7.7	9.2	4	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	8.6	7.9	9	3.7	0.7	
U2 - Toxicity prior Beta(1, 1)	8.2	7.2	9.3	4.1	1	
U3 - Toxicity prior Beta(1, 1)	8.8	7.5	9.1	3.8	0.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	3.4	1.7	0.8	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.4	1.7	0.8	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	3.3	1.7	0.8	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.3	1.6	0.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	3	1.6	0.8	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3	1.5	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.6	2.6	1.5	0.7	0.2	
U3 - Toxicity prior Beta(1, 1)	0.6	2.6	1.4	0.6	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	12.2	49	24.8	10	1.5	2.5
U3 - Toxicity prior Beta(0.15, 0.85)	14.2	52.7	22.5	7.1	1.1	2.4
U2 - Toxicity prior Beta(0.3, 0.7)	11.1	46.9	26.7	11.2	1.9	2.2
U3 - Toxicity prior Beta(0.3, 0.7)	13.6	50.4	24.6	7.8	1.5	2.1
U2 - Toxicity prior Beta(0.5, 0.5)	10	46.2	26.9	12.4	2.4	2.1
U3 - Toxicity prior Beta(0.5, 0.5)	12.6	50.9	23.2	10	1.3	2
U2 - Toxicity prior Beta(1, 1)	9.4	45.5	28	12.7	2.4	2
U3 - Toxicity prior Beta(1, 1)	11.8	51.9	23.1	9.9	1.6	1.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7.1	11.5	6.9	3.4	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	7.3	12.5	6.4	2.9	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	6.7	11.2	7.2	3.7	1	
U3 - Toxicity prior Beta(0.3, 0.7)	6.9	12	6.8	3.3	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	6.2	11.2	7.3	3.9	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	6.3	12.1	6.9	3.5	0.9	
U2 - Toxicity prior Beta(1, 1)	6	11	7.4	4.1	1.3	
U3 - Toxicity prior Beta(1, 1)	6.2	12	7	3.7	1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.9	1.7	0.9	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.9	1.6	0.9	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.8	1.7	0.9	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.8	1.6	0.9	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.5	1.5	0.9	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.4	1.5	0.9	0.2	
U2 - Toxicity prior Beta(1, 1)	0.3	3.2	1.4	0.8	0.2	
U3 - Toxicity prior Beta(1, 1)	0.4	3.1	1.4	0.8	0.2	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	8.4	7.6	19.6	49	15.2	0.2
U3 - Toxicity prior Beta(0.15, 0.85)	12	10.4	18.3	46.6	12.5	0.2
U2 - Toxicity prior Beta(0.3, 0.7)	7.5	7.6	17.1	49.4	18.3	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	10.9	10.2	17.3	48	13.5	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	7.7	7.8	15.4	50.2	18.7	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	10.5	9.1	16.9	48.1	15.2	0.2
U2 - Toxicity prior Beta(1, 1)	7.7	6.3	15.9	49.3	20.6	0.2
U3 - Toxicity prior Beta(1, 1)	11.1	9	14.7	49.3	15.7	0.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.2	4.8	6.8	10.3	3.9	
U3 - Toxicity prior Beta(0.15, 0.85)	4.5	4.9	6.9	10.4	3.3	
U2 - Toxicity prior Beta(0.3, 0.7)	4	4.6	6.7	10.5	4.2	
U3 - Toxicity prior Beta(0.3, 0.7)	4.3	4.7	6.7	10.8	3.5	
U2 - Toxicity prior Beta(0.5, 0.5)	3.9	4.4	6.5	10.8	4.4	
U3 - Toxicity prior Beta(0.5, 0.5)	4.1	4.6	6.7	10.8	3.8	
U2 - Toxicity prior Beta(1, 1)	3.8	4.3	6.3	10.9	4.7	
U3 - Toxicity prior Beta(1, 1)	4.1	4.4	6.4	11.1	4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3	1.6	2.4	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	3	1.6	2.3	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.9	1.5	2.4	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	1.5	2.3	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	1.5	2.3	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	1.5	2.2	0.7	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	1.4	2.1	0.7	
U3 - Toxicity prior Beta(1, 1)	0.2	2.8	1.5	2.1	0.6	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	5.9	33.7	25.8	23.6	10.6	0.4
U3 - Toxicity prior Beta(0.15, 0.85)	7.7	41.2	24.8	18.6	7.3	0.4
U2 - Toxicity prior Beta(0.3, 0.7)	6.6	32.2	26.2	23.7	10.9	0.4
U3 - Toxicity prior Beta(0.3, 0.7)	8.9	38.5	25.8	18.3	8.1	0.4
U2 - Toxicity prior Beta(0.5, 0.5)	5.9	30.7	27	24	12.1	0.3
U3 - Toxicity prior Beta(0.5, 0.5)	9.1	38	23.9	20.4	8.3	0.3
U2 - Toxicity prior Beta(1, 1)	5.9	29.4	27.6	24.9	11.7	0.5
U3 - Toxicity prior Beta(1, 1)	8.2	37.6	25.2	20.2	8.2	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.9	7.1	7.1	7.3	4.4	
U3 - Toxicity prior Beta(0.15, 0.85)	4.1	8.1	7.3	7	3.6	
U2 - Toxicity prior Beta(0.3, 0.7)	3.8	7	7.2	7.4	4.6	
U3 - Toxicity prior Beta(0.3, 0.7)	3.9	7.6	7.2	7.3	3.9	
U2 - Toxicity prior Beta(0.5, 0.5)	3.6	6.7	7.1	7.7	4.9	
U3 - Toxicity prior Beta(0.5, 0.5)	3.7	7.6	7.1	7.4	4.2	
U2 - Toxicity prior Beta(1, 1)	3.6	6.7	7.1	7.6	5	
U3 - Toxicity prior Beta(1, 1)	3.7	7.4	7	7.5	4.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3.2	1.9	2.4	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.2	2	2.2	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	3.1	1.9	2.3	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3.1	1.9	2.2	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.9	1.9	2.3	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	3	1.9	2.2	1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.8	1.9	2.3	1	
U3 - Toxicity prior Beta(1, 1)	0.2	2.9	1.9	2.2	0.9	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	64.2	17	3.4	1.4	0.4	13.6
U3 - Toxicity prior Beta(0.15, 0.85)	65.7	16.6	2.6	1.1	0.3	13.7
U2 - Toxicity prior Beta(0.3, 0.7)	61.7	18.7	4.6	1.7	0.4	12.9
U3 - Toxicity prior Beta(0.3, 0.7)	65.1	18.4	2.3	1.1	0.4	12.7
U2 - Toxicity prior Beta(0.5, 0.5)	60.6	19.9	5.4	2.1	0.4	11.6
U3 - Toxicity prior Beta(0.5, 0.5)	64.9	18.1	3.3	1.5	0.3	11.9
U2 - Toxicity prior Beta(1, 1)	60.4	19.7	5.7	1.8	0.4	12
U3 - Toxicity prior Beta(1, 1)	64.4	18.7	3.2	1.6	0.4	11.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.8	6.4	2.2	1	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	18.2	6.3	1.9	0.9	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	17	6.7	2.5	1.2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	17.6	6.7	2.2	1	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	16.4	6.9	3	1.2	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	17	6.9	2.6	1.1	0.3	
U2 - Toxicity prior Beta(1, 1)	16.4	6.9	2.9	1.3	0.4	
U3 - Toxicity prior Beta(1, 1)	17	6.7	2.7	1.2	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	2.6	0.6	0.3	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	2.5	0.5	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	2.5	0.6	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	2.4	0.5	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2.2	0.5	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2.2	0.5	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.6	1.8	0.5	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.7	1.7	0.5	0.2	0	

Phase I/II Operating Characteristics

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13.9	47.9	26.1	9	2.3	0.8
U3 - Toxicity prior Beta(0.15, 0.85)	15.8	54.1	22.1	6.6	0.6	0.8
U2 - Toxicity prior Beta(0.3, 0.7)	12.2	47.8	25.8	10.7	2.6	0.9
U3 - Toxicity prior Beta(0.3, 0.7)	14.8	51.3	24.5	7.8	0.8	0.8
U2 - Toxicity prior Beta(0.5, 0.5)	11.3	45.7	28.4	11	2.7	0.9
U3 - Toxicity prior Beta(0.5, 0.5)	14.1	51.2	25.3	7.8	0.9	0.7
U2 - Toxicity prior Beta(1, 1)	11.4	46.8	26.4	11.6	3	0.8
U3 - Toxicity prior Beta(1, 1)	12.6	51.7	25.6	8.5	0.9	0.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7.8	12.6	6.4	2.4	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	8.2	13.8	5.8	1.8	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	7.3	12.8	6.4	2.6	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	7.6	13.7	6.1	2	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	7	12.5	7	2.7	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	7.2	13.9	6.3	2.2	0.4	
U2 - Toxicity prior Beta(1, 1)	6.9	12.6	6.7	2.9	0.8	
U3 - Toxicity prior Beta(1, 1)	7.1	13.9	6.1	2.4	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.5	1.4	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.6	1.2	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	3.3	1.3	0.6	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3.5	1.2	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	3	1.3	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	3.1	1.2	0.5	0.1	
U2 - Toxicity prior Beta(1, 1)	0.3	2.7	1.1	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	0.3	2.7	1.1	0.4	0.1	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	46.8	11.1	2.7	1.1	0.4	37.9
U3 - Toxicity prior Beta(0.15, 0.85)	48.6	10.6	2.2	0.9	0.2	37.5
U2 - Toxicity prior Beta(0.3, 0.7)	47.8	12.6	3.5	1	0.7	34.4
U3 - Toxicity prior Beta(0.3, 0.7)	49	13.1	2.5	0.7	0.2	34.5
U2 - Toxicity prior Beta(0.5, 0.5)	45.6	16.1	3.9	0.9	0.8	32.7
U3 - Toxicity prior Beta(0.5, 0.5)	49.1	12.8	3.3	1	0.3	33.5
U2 - Toxicity prior Beta(1, 1)	46.9	13.6	4.7	0.8	1	33
U3 - Toxicity prior Beta(1, 1)	47.6	13.6	4.1	0.7	0.5	33.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	16	6.1	2	1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	16.2	5.9	2	1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	15.5	6.3	2.4	1.2	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	15.7	6.1	2.3	1.2	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	15	6.5	2.8	1.3	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	15.3	6.3	2.7	1.3	0.4	
U2 - Toxicity prior Beta(1, 1)	15	6.2	2.8	1.4	0.5	
U3 - Toxicity prior Beta(1, 1)	15.3	6.1	2.7	1.3	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	2.7	0.5	0.3	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	1	2.6	0.5	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	1	2.6	0.5	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	1	2.5	0.5	0.3	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	2.2	0.5	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	2.2	0.5	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.8	1.7	0.5	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.8	1.7	0.5	0.2	0	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.5	17.2	22.1	13.8	23.8	4.6
U3 - Toxicity prior Beta(0.15, 0.85)	24	19.4	19.2	12.9	19.7	4.8
U2 - Toxicity prior Beta(0.3, 0.7)	15.1	15.8	21.3	16.7	27	4.1
U3 - Toxicity prior Beta(0.3, 0.7)	23	17.2	19.7	13.1	22.8	4.2
U2 - Toxicity prior Beta(0.5, 0.5)	15.2	15	20.4	15.4	30.2	3.8
U3 - Toxicity prior Beta(0.5, 0.5)	21.9	15.7	20.4	13	25.2	3.8
U2 - Toxicity prior Beta(1, 1)	15.3	13.6	20.8	15.1	31.8	3.4
U3 - Toxicity prior Beta(1, 1)	18.8	17.4	19.8	13.6	27	3.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7.3	6.4	5.8	5.2	4.8	
U3 - Toxicity prior Beta(0.15, 0.85)	8	6.3	5.7	5.2	4.5	
U2 - Toxicity prior Beta(0.3, 0.7)	6.8	6.1	5.8	5.6	5.4	
U3 - Toxicity prior Beta(0.3, 0.7)	7.4	6.1	5.7	5.5	5	
U2 - Toxicity prior Beta(0.5, 0.5)	6.3	5.9	5.9	5.7	5.9	
U3 - Toxicity prior Beta(0.5, 0.5)	6.9	5.9	5.8	5.6	5.4	
U2 - Toxicity prior Beta(1, 1)	6.3	5.7	5.8	5.7	6.1	
U3 - Toxicity prior Beta(1, 1)	6.8	5.8	5.7	5.7	5.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	3.7	1.6	1.8	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.6	1.5	1.7	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.5	1.6	1.7	0.9	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.4	1.5	1.7	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.4	1.7	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.1	1.4	1.7	0.8	
U2 - Toxicity prior Beta(1, 1)	0.6	2.8	1.4	1.5	0.7	
U3 - Toxicity prior Beta(1, 1)	0.7	2.8	1.4	1.5	0.7	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.1	39.8	14.9	5.2	1.4	20.6
U3 - Toxicity prior Beta(0.15, 0.85)	19	42.2	13.4	4	0.6	20.8
U2 - Toxicity prior Beta(0.3, 0.7)	16.5	39.7	17.1	6.2	2.3	18.2
U3 - Toxicity prior Beta(0.3, 0.7)	18.7	40.8	16.4	4.1	1.5	18.5
U2 - Toxicity prior Beta(0.5, 0.5)	14.7	41.3	18.5	7.5	2.8	15.2
U3 - Toxicity prior Beta(0.5, 0.5)	16.1	45.7	15.8	5.4	1.5	15.5
U2 - Toxicity prior Beta(1, 1)	13.8	39.6	18.6	8.6	2.5	16.9
U3 - Toxicity prior Beta(1, 1)	15.5	45.1	14.8	6.4	1.5	16.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.3	10.3	4.7	2	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	10.6	10.8	4.3	1.8	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	9.8	10.4	4.8	2.4	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	10	10.8	4.7	2.1	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	9	10.5	5.5	2.6	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	9.1	11.1	5.3	2.3	0.6	
U2 - Toxicity prior Beta(1, 1)	8.9	10.3	5.5	2.7	0.9	
U3 - Toxicity prior Beta(1, 1)	9	11	5.1	2.5	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	3.9	1.1	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.9	1	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.7	1	0.6	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.7	0.9	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	0.9	0.5	0.1	
U2 - Toxicity prior Beta(1, 1)	0.5	2.7	0.9	0.5	0.1	
U3 - Toxicity prior Beta(1, 1)	0.5	2.7	0.9	0.4	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	65.4	23.7	4.7	0	0	6.2
U3 - Toxicity prior Beta(0.15, 0.85)	69.5	20.9	3.4	0	0	6.2
U2 - Toxicity prior Beta(0.3, 0.7)	65.2	24.7	4.9	0.4	0	4.8
U3 - Toxicity prior Beta(0.3, 0.7)	67.3	24.1	3.4	0.4	0.1	4.7
U2 - Toxicity prior Beta(0.5, 0.5)	64.3	25.3	5.3	0.3	0.1	4.7
U3 - Toxicity prior Beta(0.5, 0.5)	67.6	23.5	3.8	0.5	0.1	4.5
U2 - Toxicity prior Beta(1, 1)	64.7	23.4	6.3	0.6	0.1	4.9
U3 - Toxicity prior Beta(1, 1)	67.7	23	4.1	0.4	0.1	4.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	16.5	8	3.2	1.2	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	17.2	7.7	2.9	1.1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	15.7	8.1	3.7	1.5	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	16.4	7.9	3.5	1.4	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	15.1	8.2	4	1.6	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	15.8	8.1	3.7	1.5	0.3	
U2 - Toxicity prior Beta(1, 1)	15.1	8.1	4	1.7	0.4	
U3 - Toxicity prior Beta(1, 1)	15.7	8.1	3.6	1.5	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.2	3.5	0.9	0.4	0	
U3 - Toxicity prior Beta(0.15, 0.85)	1.2	3.4	0.8	0.4	0	
U2 - Toxicity prior Beta(0.3, 0.7)	1.2	3.3	0.8	0.4	0	
U3 - Toxicity prior Beta(0.3, 0.7)	1.2	3.3	0.8	0.4	0	
U2 - Toxicity prior Beta(0.5, 0.5)	1.1	2.9	0.7	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	2.9	0.7	0.3	0	
U2 - Toxicity prior Beta(1, 1)	1	2.3	0.7	0.3	0	
U3 - Toxicity prior Beta(1, 1)	1.1	2.3	0.7	0.3	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13.8	14.8	11.7	40	19.4	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	19.9	14.9	10.4	37.7	16.8	0.3
U2 - Toxicity prior Beta(0.3, 0.7)	13.3	14	10	39	23.5	0.2
U3 - Toxicity prior Beta(0.3, 0.7)	18.4	14.3	8.6	38.1	20.3	0.3
U2 - Toxicity prior Beta(0.5, 0.5)	12.1	13	9	40	25.6	0.3
U3 - Toxicity prior Beta(0.5, 0.5)	17.8	13.4	8.3	37.9	22.3	0.3
U2 - Toxicity prior Beta(1, 1)	12.9	12.5	8.9	38.3	27.2	0.2
U3 - Toxicity prior Beta(1, 1)	17.4	13.4	8.6	36.6	23.8	0.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.7	5.4	5.7	9	5.3	
U3 - Toxicity prior Beta(0.15, 0.85)	5.1	5.5	5.4	9.2	4.7	
U2 - Toxicity prior Beta(0.3, 0.7)	4.4	5.1	5.5	9.1	5.8	
U3 - Toxicity prior Beta(0.3, 0.7)	4.8	5.2	5.3	9.4	5.2	
U2 - Toxicity prior Beta(0.5, 0.5)	4.3	4.9	5.2	9.3	6.2	
U3 - Toxicity prior Beta(0.5, 0.5)	4.5	5.1	5.1	9.6	5.6	
U2 - Toxicity prior Beta(1, 1)	4.1	4.8	5.2	9.3	6.6	
U3 - Toxicity prior Beta(1, 1)	4.5	4.9	5	9.6	6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.1	1.6	2.3	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.2	1.6	2.3	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	3.1	1.6	2.3	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3.1	1.5	2.3	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.9	1.5	2.2	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3	1.5	2.2	1.1	
U2 - Toxicity prior Beta(1, 1)	0.3	2.8	1.5	2.1	1	
U3 - Toxicity prior Beta(1, 1)	0.4	2.8	1.5	2.1	0.9	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	20.1	40.5	13.8	6.3	1.6	17.7
U3 - Toxicity prior Beta(0.15, 0.85)	22.9	42.5	11.8	5.1	1	16.7
U2 - Toxicity prior Beta(0.3, 0.7)	17.9	41.3	16.5	7.5	1.8	15
U3 - Toxicity prior Beta(0.3, 0.7)	21.3	42	14.8	5.8	1.2	14.9
U2 - Toxicity prior Beta(0.5, 0.5)	16.3	42.7	19.8	7.3	1.7	12.2
U3 - Toxicity prior Beta(0.5, 0.5)	19.9	43.4	16.5	5.8	1.4	13
U2 - Toxicity prior Beta(1, 1)	17.1	41.8	19.4	7	2.2	12.5
U3 - Toxicity prior Beta(1, 1)	19.1	44.2	16.1	5.6	1.6	13.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.2	9.7	4.8	2.8	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	10.6	10	4.6	2.5	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	9.6	9.6	5.1	3.2	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	9.8	9.8	5	3	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	8.7	9.7	5.8	3.4	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	8.9	9.9	5.5	3.3	1.1	
U2 - Toxicity prior Beta(1, 1)	8.8	9.5	5.7	3.5	1.3	
U3 - Toxicity prior Beta(1, 1)	9	9.8	5.6	3.3	1.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	4	1.2	0.9	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	4.1	1.1	0.8	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	3.8	1.1	0.8	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.8	1.1	0.7	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	3.4	1.1	0.8	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.4	1	0.8	0.1	
U2 - Toxicity prior Beta(1, 1)	0.5	2.8	1	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	0.5	2.9	1	0.6	0.1	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.8	18.6	15.9	28.8	12.9	1
U3 - Toxicity prior Beta(0.15, 0.85)	28.8	20.7	13.2	27.1	9.1	1.1
U2 - Toxicity prior Beta(0.3, 0.7)	22.3	17.3	14.9	28.4	16.3	0.8
U3 - Toxicity prior Beta(0.3, 0.7)	28.2	20.2	12.9	26.8	11.1	0.8
U2 - Toxicity prior Beta(0.5, 0.5)	19.1	17.4	14.5	29.9	18.3	0.8
U3 - Toxicity prior Beta(0.5, 0.5)	29	18.5	12	28.1	11.6	0.8
U2 - Toxicity prior Beta(1, 1)	17.1	16.8	14.5	31.9	18.9	0.8
U3 - Toxicity prior Beta(1, 1)	27	19.4	11.6	28.2	13	0.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.2	6.3	5.9	7.3	4.2	
U3 - Toxicity prior Beta(0.15, 0.85)	6.8	6.4	5.6	7.2	3.9	
U2 - Toxicity prior Beta(0.3, 0.7)	5.7	5.9	5.8	7.6	4.8	
U3 - Toxicity prior Beta(0.3, 0.7)	6.4	6.1	5.5	7.5	4.5	
U2 - Toxicity prior Beta(0.5, 0.5)	5.3	5.8	5.8	7.9	5.2	
U3 - Toxicity prior Beta(0.5, 0.5)	5.8	5.9	5.5	7.9	4.8	
U2 - Toxicity prior Beta(1, 1)	5.2	5.6	5.7	8	5.4	
U3 - Toxicity prior Beta(1, 1)	5.8	5.7	5.4	8.1	5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.6	1.7	2.2	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	3.6	1.7	2.1	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.4	1.6	2.2	0.9	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.5	1.5	2.1	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.5	2.2	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.3	1.5	2.1	0.8	
U2 - Toxicity prior Beta(1, 1)	0.6	3	1.5	2	0.8	
U3 - Toxicity prior Beta(1, 1)	0.7	3	1.5	2	0.7	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	50.3	30.3	13.2	3.8	0.6	1.8
U3 - Toxicity prior Beta(0.15, 0.85)	55.3	29.6	9.9	2.6	0.5	2.1
U2 - Toxicity prior Beta(0.3, 0.7)	48.6	31	13.7	4	0.8	1.9
U3 - Toxicity prior Beta(0.3, 0.7)	54.6	30.4	10.3	2.3	0.4	2
U2 - Toxicity prior Beta(0.5, 0.5)	49.2	30.8	12.8	4.6	1.2	1.4
U3 - Toxicity prior Beta(0.5, 0.5)	57.3	28.2	9.7	2.6	0.5	1.7
U2 - Toxicity prior Beta(1, 1)	48.4	31.8	12.4	5.1	0.9	1.4
U3 - Toxicity prior Beta(1, 1)	57.2	28.2	9.3	3.5	0.4	1.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.1	8.7	5.7	3.9	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	11	8.4	5.3	3.8	1.3	
U2 - Toxicity prior Beta(0.3, 0.7)	9.6	8.5	5.9	4.2	1.6	
U3 - Toxicity prior Beta(0.3, 0.7)	10.5	8.4	5.4	4	1.5	
U2 - Toxicity prior Beta(0.5, 0.5)	9.2	8.4	6	4.4	1.8	
U3 - Toxicity prior Beta(0.5, 0.5)	10	8.2	5.6	4.2	1.7	
U2 - Toxicity prior Beta(1, 1)	8.9	8.3	6.1	4.6	1.9	
U3 - Toxicity prior Beta(1, 1)	10	8.1	5.5	4.4	1.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.2	4.2	1.8	1.4	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	1.3	4.1	1.7	1.4	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	1.2	4.1	1.8	1.4	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	1.3	4	1.6	1.4	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	1.1	3.9	1.6	1.4	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	1.3	3.8	1.5	1.4	0.3	
U2 - Toxicity prior Beta(1, 1)	1.1	3.6	1.6	1.4	0.3	
U3 - Toxicity prior Beta(1, 1)	1.3	3.5	1.5	1.3	0.3	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	38.2	25.3	16.7	11.8	8	0
U3 - Toxicity prior Beta(0.15, 0.85)	43.6	28.7	14	9.4	4.3	0
U2 - Toxicity prior Beta(0.3, 0.7)	38.2	26.4	15.7	12.3	7.4	0
U3 - Toxicity prior Beta(0.3, 0.7)	42.9	27.6	15.6	8.7	5.2	0
U2 - Toxicity prior Beta(0.5, 0.5)	37.3	26.1	16.8	11.4	8.4	0
U3 - Toxicity prior Beta(0.5, 0.5)	42.9	27.7	15.6	9.1	4.7	0
U2 - Toxicity prior Beta(1, 1)	38.4	26.8	17	10.7	7.1	0
U3 - Toxicity prior Beta(1, 1)	42.9	28.3	14.9	9.8	4.1	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.3	5	6.1	8	6.5	
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	5.5	6.3	8	5.6	
U2 - Toxicity prior Beta(0.3, 0.7)	4.3	5	6.1	8.2	6.4	
U3 - Toxicity prior Beta(0.3, 0.7)	4.6	5.4	6.3	7.9	5.8	
U2 - Toxicity prior Beta(0.5, 0.5)	4.3	4.9	6	8.1	6.6	
U3 - Toxicity prior Beta(0.5, 0.5)	4.5	5.3	6.2	8.2	5.9	
U2 - Toxicity prior Beta(1, 1)	4.3	4.9	6	8.1	6.7	
U3 - Toxicity prior Beta(1, 1)	4.5	5.2	6.2	8.1	5.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	2.2	2.9	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	2.3	2.9	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	2.2	3	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	2.2	2.9	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	2.1	3	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.9	2.2	2.9	1.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	2.1	3	1.4	
U3 - Toxicity prior Beta(1, 1)	0.3	2.8	2.2	2.9	1.3	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.1	59.8	16.8	1.2	0	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	21.8	62.9	13.9	1.3	0	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	19.8	60.7	17.8	1.6	0	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	21.7	62.7	14.5	1	0	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	19	59.5	19.2	2.2	0	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	21.5	61	15.9	1.3	0.1	0.2
U2 - Toxicity prior Beta(1, 1)	18.8	60.1	18.9	2.1	0	0.1
U3 - Toxicity prior Beta(1, 1)	21.2	62.2	14.7	1.8	0	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.3	14.6	5.1	0.9	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	9.5	15.3	4.4	0.7	0	
U2 - Toxicity prior Beta(0.3, 0.7)	8.8	14.5	5.5	1.1	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	9.1	15.2	4.9	0.8	0	
U2 - Toxicity prior Beta(0.5, 0.5)	8.3	14.5	5.8	1.2	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	8.8	15.1	5	0.9	0.1	
U2 - Toxicity prior Beta(1, 1)	8.2	14.6	5.8	1.3	0.1	
U3 - Toxicity prior Beta(1, 1)	8.7	15.2	5.1	0.9	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.9	1.3	0.3	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.9	1.1	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.7	1.3	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.8	1.1	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.5	1.2	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.5	1	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.5	3.2	1	0.3	0	
U3 - Toxicity prior Beta(1, 1)	0.5	3.3	0.9	0.2	0	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	12.2	10.2	10.4	16.8	50.4	0
U3 - Toxicity prior Beta(0.15, 0.85)	18.6	12.5	9.6	13.8	45.5	0
U2 - Toxicity prior Beta(0.3, 0.7)	12.2	8.5	10.5	15.9	52.9	0
U3 - Toxicity prior Beta(0.3, 0.7)	20.1	12.1	9.4	13	45.4	0
U2 - Toxicity prior Beta(0.5, 0.5)	11.6	8.5	10	14	55.9	0
U3 - Toxicity prior Beta(0.5, 0.5)	18.8	11	9.9	13.2	47.1	0
U2 - Toxicity prior Beta(1, 1)	10.4	8.3	10.8	13.9	56.6	0
U3 - Toxicity prior Beta(1, 1)	17.1	12.1	9.3	12.9	48.6	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.4	3.9	5.2	6.9	10.5	
U3 - Toxicity prior Beta(0.15, 0.85)	3.6	4.2	5.2	7	10.1	
U2 - Toxicity prior Beta(0.3, 0.7)	3.4	3.9	5	6.8	10.9	
U3 - Toxicity prior Beta(0.3, 0.7)	3.5	4.1	5.1	7	10.3	
U2 - Toxicity prior Beta(0.5, 0.5)	3.3	3.8	5	6.8	11.1	
U3 - Toxicity prior Beta(0.5, 0.5)	3.5	4	5.1	7	10.4	
U2 - Toxicity prior Beta(1, 1)	3.3	3.7	4.9	6.7	11.4	
U3 - Toxicity prior Beta(1, 1)	3.4	3.9	4.9	7	10.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.6	1.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.6	2.5	1.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.5	1.6	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1.5	2.5	1.6	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.5	2.5	1.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.6	2.5	1.6	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.5	2.4	1.5	
U3 - Toxicity prior Beta(1, 1)	0.1	2.7	1.5	2.5	1.5	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	20.7	57.8	17.6	1.4	0.1	2.4
U3 - Toxicity prior Beta(0.15, 0.85)	23.5	58.7	14.5	0.6	0.1	2.6
U2 - Toxicity prior Beta(0.3, 0.7)	18.8	59.3	18.3	1.7	0	1.9
U3 - Toxicity prior Beta(0.3, 0.7)	20.8	60.3	14.9	1.8	0	2.2
U2 - Toxicity prior Beta(0.5, 0.5)	19.2	58.4	18.4	1.8	0.1	2.1
U3 - Toxicity prior Beta(0.5, 0.5)	22.7	57.5	15	2.5	0.1	2.2
U2 - Toxicity prior Beta(1, 1)	19.5	56.7	19.3	2.5	0	2
U3 - Toxicity prior Beta(1, 1)	21.9	57.7	16	2.7	0	1.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.2	12.5	6.5	2.3	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	8.6	12.7	6.1	2.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	7.8	12.2	6.8	2.6	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	8.1	12.6	6.4	2.4	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	7.5	12.1	7	2.8	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	7.8	12.4	6.7	2.6	0.4	
U2 - Toxicity prior Beta(1, 1)	7.3	12	7.1	2.9	0.6	
U3 - Toxicity prior Beta(1, 1)	7.6	12.4	6.8	2.7	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	4.6	1.7	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	4.6	1.6	0.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	4.5	1.6	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	4.5	1.5	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	4	1.4	0.7	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	3.9	1.3	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	3.5	1.2	0.6	0	
U3 - Toxicity prior Beta(1, 1)	0.7	3.5	1.2	0.6	0	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	12.3	49.3	26.9	9.7	1.5	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	14.9	52.1	23.2	8.8	0.7	0.3
U2 - Toxicity prior Beta(0.3, 0.7)	11.7	47.4	28.5	10.3	1.8	0.3
U3 - Toxicity prior Beta(0.3, 0.7)	16.1	49.5	25.2	7.7	1.2	0.3
U2 - Toxicity prior Beta(0.5, 0.5)	12.5	44.9	29.1	10.9	2.2	0.4
U3 - Toxicity prior Beta(0.5, 0.5)	17.5	47.4	23.2	10.2	1.4	0.3
U2 - Toxicity prior Beta(1, 1)	13.1	44.2	28.1	11.8	2.3	0.5
U3 - Toxicity prior Beta(1, 1)	18.9	46.3	24	8.5	1.7	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.9	10	8.2	5.1	1.7	
U3 - Toxicity prior Beta(0.15, 0.85)	5.3	10.5	7.8	4.8	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	4.7	9.6	8.3	5.3	2	
U3 - Toxicity prior Beta(0.3, 0.7)	5	10.2	7.9	5	1.7	
U2 - Toxicity prior Beta(0.5, 0.5)	4.5	9.3	8.3	5.6	2.2	
U3 - Toxicity prior Beta(0.5, 0.5)	4.9	9.9	7.9	5.3	2	
U2 - Toxicity prior Beta(1, 1)	4.4	8.9	8.2	5.9	2.5	
U3 - Toxicity prior Beta(1, 1)	4.8	9.7	7.7	5.5	2.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.9	2.3	1.7	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.9	2.2	1.6	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.7	2.2	1.7	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.8	2.1	1.6	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.6	2.1	1.7	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.6	2.1	1.6	0.4	
U2 - Toxicity prior Beta(1, 1)	0.4	3.4	2	1.7	0.4	
U3 - Toxicity prior Beta(1, 1)	0.5	3.5	2	1.5	0.4	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	54.9	2.1	0	0	0	43
U3 - Toxicity prior Beta(0.15, 0.85)	55	1.9	0	0	0	43.1
U2 - Toxicity prior Beta(0.3, 0.7)	56.2	2.3	0	0	0	41.5
U3 - Toxicity prior Beta(0.3, 0.7)	56.5	1.9	0.1	0	0	41.5
U2 - Toxicity prior Beta(0.5, 0.5)	57.6	2.3	0	0	0	40.1
U3 - Toxicity prior Beta(0.5, 0.5)	57	2.5	0	0	0	40.5
U2 - Toxicity prior Beta(1, 1)	56.7	2.6	0	0	0	40.7
U3 - Toxicity prior Beta(1, 1)	56.7	2.2	0	0	0	41.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	19.7	2.8	0.2	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	19.7	2.8	0.1	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	19.7	3	0.2	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	19.8	2.9	0.2	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	19.8	3.1	0.3	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	19.9	3	0.2	0	0	
U2 - Toxicity prior Beta(1, 1)	19.7	3.1	0.3	0	0	
U3 - Toxicity prior Beta(1, 1)	19.9	3	0.2	0	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	1.1	0	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	1.1	0	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	1	0	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	1	0	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	0.9	0	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	0.9	0	0	0	
U2 - Toxicity prior Beta(1, 1)	0.1	0.7	0	0	0	
U3 - Toxicity prior Beta(1, 1)	0.1	0.7	0	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	23.5	59.9	12.7	2.2	0	1.7
U3 - Toxicity prior Beta(0.15, 0.85)	27.7	55.6	13	2	0	1.7
U2 - Toxicity prior Beta(0.3, 0.7)	23	57.9	14.5	3.1	0.1	1.4
U3 - Toxicity prior Beta(0.3, 0.7)	27.4	55.2	13.5	2.5	0	1.4
U2 - Toxicity prior Beta(0.5, 0.5)	22.8	59	13.7	3.2	0.1	1.2
U3 - Toxicity prior Beta(0.5, 0.5)	26.7	56.3	12.8	3	0	1.2
U2 - Toxicity prior Beta(1, 1)	22.7	57.8	14.2	4.1	0.1	1.1
U3 - Toxicity prior Beta(1, 1)	26.6	57.1	11.5	3.5	0	1.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.9	12.8	5.8	2.3	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	9.3	12.7	5.6	2.1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	8.5	12.5	6.2	2.5	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	9	12.4	5.9	2.4	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	8.2	12.4	6.2	2.8	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	8.7	12.4	6	2.6	0.3	
U2 - Toxicity prior Beta(1, 1)	8	12.2	6.3	3	0.4	
U3 - Toxicity prior Beta(1, 1)	8.5	12.2	6	2.8	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	4.7	1.6	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	1.2	4.6	1.6	0.6	0	
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	4.5	1.5	0.6	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	1.2	4.4	1.5	0.6	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	1	4.2	1.3	0.6	0	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	4.1	1.3	0.5	0	
U2 - Toxicity prior Beta(1, 1)	0.9	3.7	1.1	0.5	0	
U3 - Toxicity prior Beta(1, 1)	1	3.7	1.1	0.5	0	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	20.9	4	0.1	0	0	75
U3 - Toxicity prior Beta(0.15, 0.85)	20.5	3.8	0.1	0	0	75.6
U2 - Toxicity prior Beta(0.3, 0.7)	21.8	3.9	0	0	0	74.3
U3 - Toxicity prior Beta(0.3, 0.7)	20.8	4.1	0.1	0	0	75
U2 - Toxicity prior Beta(0.5, 0.5)	22.4	4	0	0	0	73.6
U3 - Toxicity prior Beta(0.5, 0.5)	21.3	4.5	0.3	0	0	73.9
U2 - Toxicity prior Beta(1, 1)	21.8	4	0.1	0	0	74.1
U3 - Toxicity prior Beta(1, 1)	21.1	4.5	0	0	0	74.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	15.2	5	1	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	15.2	4.8	1	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	15.2	5.1	1.1	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	15.3	4.9	1.1	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	15.3	5.2	1.3	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	15.4	5.1	1.3	0.2	0	
U2 - Toxicity prior Beta(1, 1)	15.4	5	1.2	0.2	0	
U3 - Toxicity prior Beta(1, 1)	15.4	4.9	1.2	0.2	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	2.3	0.2	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	2.3	0.2	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	2	0.2	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	2	0.2	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	1.8	0.2	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	1.8	0.2	0	0	
U2 - Toxicity prior Beta(1, 1)	0.4	1.4	0.2	0	0	
U3 - Toxicity prior Beta(1, 1)	0.4	1.4	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	1.6	0.9	3.9	49.6	44	0
U3 - Toxicity prior Beta(0.15, 0.85)	3.4	1.4	4.1	52.5	38.6	0
U2 - Toxicity prior Beta(0.3, 0.7)	1.4	1.1	3.5	49.3	44.7	0
U3 - Toxicity prior Beta(0.3, 0.7)	2.9	1.2	4.3	52.8	38.8	0
U2 - Toxicity prior Beta(0.5, 0.5)	1.3	0.8	4.3	48.2	45.4	0
U3 - Toxicity prior Beta(0.5, 0.5)	3.1	1.4	4.9	51.9	38.7	0
U2 - Toxicity prior Beta(1, 1)	1.5	1.2	3.4	48.8	45.1	0
U3 - Toxicity prior Beta(1, 1)	2.9	1.3	4.2	52.9	38.7	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.1	3.1	3.9	11.8	8.1	
U3 - Toxicity prior Beta(0.15, 0.85)	3.1	3.2	3.9	12.5	7.4	
U2 - Toxicity prior Beta(0.3, 0.7)	3.1	3.1	3.8	11.8	8.2	
U3 - Toxicity prior Beta(0.3, 0.7)	3.1	3.2	3.9	12.4	7.5	
U2 - Toxicity prior Beta(0.5, 0.5)	3.1	3.1	3.8	11.7	8.3	
U3 - Toxicity prior Beta(0.5, 0.5)	3.1	3.2	3.9	12.3	7.5	
U2 - Toxicity prior Beta(1, 1)	3.1	3.1	3.8	11.8	8.2	
U3 - Toxicity prior Beta(1, 1)	3.1	3.1	3.8	12.4	7.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0	2.4	1.1	2.5	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0	2.4	1.1	2.5	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0	2.4	1	2.5	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0	2.4	1.1	2.5	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0	2.4	1	2.5	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0	2.4	1.1	2.5	1.5	
U2 - Toxicity prior Beta(1, 1)	0	2.4	1.1	2.4	1.5	
U3 - Toxicity prior Beta(1, 1)	0	2.4	1.1	2.5	1.5	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	29.3	35	30.1	5.5	0	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	34.6	34.2	26.7	4.4	0	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	28.6	34.4	31.3	5.5	0.1	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	34.7	33	27.9	4.3	0	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	28.4	31.6	33.3	6.5	0.1	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	33.5	31.9	29.1	5.4	0	0.1
U2 - Toxicity prior Beta(1, 1)	26.3	34.5	32.2	6.6	0.2	0.2
U3 - Toxicity prior Beta(1, 1)	33.2	33.2	27.2	6.1	0.2	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.8	9.9	8.4	2.6	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	9.3	10	8.1	2.4	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	8.4	9.6	8.7	2.9	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	8.8	9.7	8.5	2.7	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	7.8	9.5	9.1	3.2	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	8.4	9.6	8.7	2.9	0.3	
U2 - Toxicity prior Beta(1, 1)	7.9	9.2	9.1	3.4	0.5	
U3 - Toxicity prior Beta(1, 1)	8.4	9.4	8.8	3	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	3.8	1.9	0.9	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	3.8	1.9	0.9	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	3.7	1.9	0.9	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	3.7	1.8	0.9	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	3.5	1.7	0.9	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	3.5	1.7	0.8	0.1	
U2 - Toxicity prior Beta(1, 1)	0.8	3.2	1.6	0.8	0.1	
U3 - Toxicity prior Beta(1, 1)	0.9	3.3	1.5	0.7	0	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	27.1	35.5	29.1	5	0.1	3.2
U3 - Toxicity prior Beta(0.15, 0.85)	33.4	34	25	4.4	0.2	3
U2 - Toxicity prior Beta(0.3, 0.7)	25.5	35.2	30.8	5.7	0.1	2.7
U3 - Toxicity prior Beta(0.3, 0.7)	30.3	33.9	27.5	5.4	0.1	2.8
U2 - Toxicity prior Beta(0.5, 0.5)	24.6	33.9	32	6.3	0	3.2
U3 - Toxicity prior Beta(0.5, 0.5)	27.6	35.9	27.9	5.8	0	2.8
U2 - Toxicity prior Beta(1, 1)	23.9	35	31.9	6.4	0	2.8
U3 - Toxicity prior Beta(1, 1)	30.1	33.9	28.1	5.2	0	2.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9	9.5	7.9	3	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	9.5	9.4	7.6	2.9	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	8.6	9.1	8.2	3.4	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	9	9.2	7.9	3.2	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	8	9.1	8.5	3.6	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	8.5	9.3	8.2	3.4	0.4	
U2 - Toxicity prior Beta(1, 1)	7.9	9	8.6	3.7	0.6	
U3 - Toxicity prior Beta(1, 1)	8.5	8.9	8.3	3.5	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	4.1	1.7	1.1	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	4	1.6	1.1	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	4	1.6	1.1	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	3.9	1.5	1	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	3.5	1.4	1	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	3.6	1.3	1	0.1	
U2 - Toxicity prior Beta(1, 1)	0.8	3.2	1.3	0.8	0.1	
U3 - Toxicity prior Beta(1, 1)	0.9	3.2	1.2	0.8	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	19.1	30.5	39.6	9.5	0.5	0.8
U3 - Toxicity prior Beta(0.15, 0.85)	23.1	30.5	36.6	8.7	0.3	0.8
U2 - Toxicity prior Beta(0.3, 0.7)	16.4	29.7	42.4	10.2	0.5	0.8
U3 - Toxicity prior Beta(0.3, 0.7)	21.3	29.7	39.1	8.6	0.5	0.8
U2 - Toxicity prior Beta(0.5, 0.5)	15.8	28.1	43.5	11.2	0.9	0.5
U3 - Toxicity prior Beta(0.5, 0.5)	19.7	28.9	39.8	10.6	0.5	0.5
U2 - Toxicity prior Beta(1, 1)	14.5	30.7	41.8	11.4	1.2	0.4
U3 - Toxicity prior Beta(1, 1)	19.3	30.3	40.2	9.3	0.5	0.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.8	8.6	9.8	4	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	7.2	8.7	9.6	3.7	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	6.3	8.3	10.1	4.4	0.9	
U3 - Toxicity prior Beta(0.3, 0.7)	6.8	8.4	9.9	4.1	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	6	8	10.2	4.6	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	6.3	8.3	10	4.4	0.9	
U2 - Toxicity prior Beta(1, 1)	5.9	8	10	4.9	1.1	
U3 - Toxicity prior Beta(1, 1)	6.3	8.1	10.1	4.6	0.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.8	2.1	1.4	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.7	2	1.3	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.6	2	1.3	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.7	2	1.2	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.4	1.9	1.3	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.4	1.8	1.2	0.2	
U2 - Toxicity prior Beta(1, 1)	0.5	3.1	1.7	1.1	0.2	
U3 - Toxicity prior Beta(1, 1)	0.6	3.2	1.7	1.1	0.1	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	29	46.9	5.1	1	0	18
U3 - Toxicity prior Beta(0.15, 0.85)	30.2	45.8	5.4	0.3	0	18.3
U2 - Toxicity prior Beta(0.3, 0.7)	26.9	48.4	7.5	0.7	0	16.5
U3 - Toxicity prior Beta(0.3, 0.7)	29.5	46.9	6.6	0.5	0	16.5
U2 - Toxicity prior Beta(0.5, 0.5)	25.1	50.3	8.7	0.4	0.1	15.4
U3 - Toxicity prior Beta(0.5, 0.5)	26.3	50.6	7.5	0.2	0	15.4
U2 - Toxicity prior Beta(1, 1)	25.8	50	8.3	1.1	0	14.8
U3 - Toxicity prior Beta(1, 1)	26.2	49.9	7.7	0.7	0	15.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.8	11.4	3.9	1.1	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	12.2	11.5	3.6	1	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	11.1	11.6	4.2	1.3	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	11.5	11.6	4	1.3	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	10.5	11.8	4.6	1.4	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	10.9	11.9	4.4	1.3	0.2	
U2 - Toxicity prior Beta(1, 1)	10.5	11.9	4.5	1.5	0.2	
U3 - Toxicity prior Beta(1, 1)	10.6	11.9	4.5	1.4	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	4.4	1.1	0.4	0	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	4.4	1	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	1	4.1	1	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	4.1	0.9	0.3	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.9	3.6	0.9	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	3.6	0.9	0.3	0	
U2 - Toxicity prior Beta(1, 1)	0.7	3	0.8	0.3	0	
U3 - Toxicity prior Beta(1, 1)	0.8	3	0.7	0.2	0	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	60.9	31.9	4.8	0.1	0	2.3
U3 - Toxicity prior Beta(0.15, 0.85)	63.5	30.5	3.6	0	0	2.4
U2 - Toxicity prior Beta(0.3, 0.7)	58.8	33.9	4.7	0.2	0	2.4
U3 - Toxicity prior Beta(0.3, 0.7)	62.3	31.7	3.6	0.1	0	2.3
U2 - Toxicity prior Beta(0.5, 0.5)	57.6	33.2	6.5	0.3	0	2.4
U3 - Toxicity prior Beta(0.5, 0.5)	62.6	31.3	3.9	0.2	0	2
U2 - Toxicity prior Beta(1, 1)	58	34.5	5	0.3	0	2.2
U3 - Toxicity prior Beta(1, 1)	60.6	33	4	0.2	0	2.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.4	9.4	2.3	0.4	0	
U3 - Toxicity prior Beta(0.15, 0.85)	18.1	9.2	1.9	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	16.7	9.8	2.6	0.5	0	
U3 - Toxicity prior Beta(0.3, 0.7)	17.5	9.5	2.1	0.4	0	
U2 - Toxicity prior Beta(0.5, 0.5)	16.2	9.9	2.9	0.5	0	
U3 - Toxicity prior Beta(0.5, 0.5)	16.9	9.8	2.4	0.4	0	
U2 - Toxicity prior Beta(1, 1)	16	10.1	3	0.6	0	
U3 - Toxicity prior Beta(1, 1)	16.8	9.8	2.4	0.5	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	3.2	0.8	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	3	0.7	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	3.1	0.8	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	3	0.7	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	2.8	0.7	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	2.7	0.6	0.1	0	
U2 - Toxicity prior Beta(1, 1)	0.7	2.4	0.6	0.1	0	
U3 - Toxicity prior Beta(1, 1)	0.7	2.3	0.5	0.1	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	19.2	15.3	13.7	39.6	12.2	0
U3 - Toxicity prior Beta(0.15, 0.85)	26.1	16.9	12.5	34.1	10.4	0
U2 - Toxicity prior Beta(0.3, 0.7)	19.9	13.1	14.2	39.6	13.2	0
U3 - Toxicity prior Beta(0.3, 0.7)	24.9	16.2	13.2	35.3	10.4	0
U2 - Toxicity prior Beta(0.5, 0.5)	16.6	12.9	14.4	40.7	15.4	0
U3 - Toxicity prior Beta(0.5, 0.5)	23.8	15.9	11.1	36.8	12.4	0
U2 - Toxicity prior Beta(1, 1)	17.1	13	13.2	41.4	15.3	0
U3 - Toxicity prior Beta(1, 1)	24.2	16.6	11.1	36.5	11.6	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.4	5	6.3	10.1	4.3	
U3 - Toxicity prior Beta(0.15, 0.85)	4.7	5.3	6.2	9.9	3.9	
U2 - Toxicity prior Beta(0.3, 0.7)	4.3	4.8	6.1	10.1	4.7	
U3 - Toxicity prior Beta(0.3, 0.7)	4.5	5.2	6	10.1	4.2	
U2 - Toxicity prior Beta(0.5, 0.5)	4.1	4.7	5.9	10.2	5	
U3 - Toxicity prior Beta(0.5, 0.5)	4.3	5	5.8	10.2	4.6	
U2 - Toxicity prior Beta(1, 1)	4.1	4.7	5.7	10.2	5.3	
U3 - Toxicity prior Beta(1, 1)	4.3	5	5.7	10.2	4.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.8	1.8	2.4	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	1.9	2.3	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1.8	2.3	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.8	1.8	2.3	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.7	2.3	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	1.7	2.2	1.1	
U2 - Toxicity prior Beta(1, 1)	0.2	2.6	1.6	2.2	1.1	
U3 - Toxicity prior Beta(1, 1)	0.3	2.7	1.7	2.1	1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	3.1	33.3	29	27.9	6.7	0
U3 - Toxicity prior Beta(0.15, 0.85)	5	38.9	29.5	22.9	3.7	0
U2 - Toxicity prior Beta(0.3, 0.7)	3	31.9	31.1	28.8	5.2	0
U3 - Toxicity prior Beta(0.3, 0.7)	4.8	38.8	29.6	24.1	2.7	0
U2 - Toxicity prior Beta(0.5, 0.5)	3.6	31.8	29.6	29	6	0
U3 - Toxicity prior Beta(0.5, 0.5)	4.8	38	30.5	23.3	3.4	0
U2 - Toxicity prior Beta(1, 1)	3.3	32.2	29.8	28.7	6	0
U3 - Toxicity prior Beta(1, 1)	5	37.8	30.1	23.9	3.2	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.8	11.1	8.1	5.5	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	4	12.2	8	4.8	1	
U2 - Toxicity prior Beta(0.3, 0.7)	3.8	11.1	8	5.5	1.6	
U3 - Toxicity prior Beta(0.3, 0.7)	3.9	12.1	8	5	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	3.8	10.8	7.9	5.8	1.7	
U3 - Toxicity prior Beta(0.5, 0.5)	3.9	12.1	8.1	5	1	
U2 - Toxicity prior Beta(1, 1)	3.8	10.8	8	5.8	1.6	
U3 - Toxicity prior Beta(1, 1)	3.9	12	8	5.1	1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.6	1.6	1.3	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.8	1.6	1.1	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0	2.6	1.7	1.3	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.8	1.6	1.1	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0	2.6	1.6	1.4	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.7	1.1	0.3	
U2 - Toxicity prior Beta(1, 1)	0	2.6	1.7	1.3	0.5	
U3 - Toxicity prior Beta(1, 1)	0.1	2.7	1.7	1.1	0.3	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13.4	13.7	31.8	28.6	12.5	0
U3 - Toxicity prior Beta(0.15, 0.85)	18.6	18.1	31.4	24.7	7.2	0
U2 - Toxicity prior Beta(0.3, 0.7)	13.2	13.5	32.3	27.5	13.5	0
U3 - Toxicity prior Beta(0.3, 0.7)	18.2	18.3	32.3	23.7	7.5	0
U2 - Toxicity prior Beta(0.5, 0.5)	13.6	13.3	30.8	28.1	14.2	0
U3 - Toxicity prior Beta(0.5, 0.5)	18.3	17.6	31.6	23.8	8.7	0
U2 - Toxicity prior Beta(1, 1)	12.6	13.4	31.5	28.3	14.2	0
U3 - Toxicity prior Beta(1, 1)	18.2	16.8	32.4	24	8.6	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.5	4.2	8.3	8.9	5.1	
U3 - Toxicity prior Beta(0.15, 0.85)	3.7	4.6	8.5	8.8	4.4	
U2 - Toxicity prior Beta(0.3, 0.7)	3.5	4.1	8.2	8.9	5.4	
U3 - Toxicity prior Beta(0.3, 0.7)	3.6	4.5	8.5	8.8	4.6	
U2 - Toxicity prior Beta(0.5, 0.5)	3.4	4.1	7.9	8.9	5.6	
U3 - Toxicity prior Beta(0.5, 0.5)	3.5	4.4	8.3	8.9	4.9	
U2 - Toxicity prior Beta(1, 1)	3.4	4	7.8	9	5.8	
U3 - Toxicity prior Beta(1, 1)	3.4	4.3	8.1	9.1	5.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.9	2.9	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	2	2.8	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.9	2.9	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	2	2.8	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.8	2.9	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	2	2.8	1.1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.8	2.9	1.1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.7	1.9	2.8	1.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9	6.1	13.5	43	28.4	0
U3 - Toxicity prior Beta(0.15, 0.85)	14.5	7.2	13	40.6	24.7	0
U2 - Toxicity prior Beta(0.3, 0.7)	9.1	5.6	12.7	41.6	31	0
U3 - Toxicity prior Beta(0.3, 0.7)	13.9	7.5	13	39.5	26.1	0
U2 - Toxicity prior Beta(0.5, 0.5)	9.3	4.3	12.6	41.2	32.6	0
U3 - Toxicity prior Beta(0.5, 0.5)	13.8	7.5	13	38.7	27	0
U2 - Toxicity prior Beta(1, 1)	8.4	4.2	12.7	41.2	33.5	0
U3 - Toxicity prior Beta(1, 1)	12.6	7.1	11.8	39.6	28.9	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.4	3.7	5.9	10.2	6.9	
U3 - Toxicity prior Beta(0.15, 0.85)	3.5	3.9	5.9	10.5	6.2	
U2 - Toxicity prior Beta(0.3, 0.7)	3.3	3.7	5.8	10.1	7	
U3 - Toxicity prior Beta(0.3, 0.7)	3.5	3.8	5.8	10.3	6.5	
U2 - Toxicity prior Beta(0.5, 0.5)	3.3	3.6	5.7	10.1	7.3	
U3 - Toxicity prior Beta(0.5, 0.5)	3.4	3.7	5.7	10.5	6.7	
U2 - Toxicity prior Beta(1, 1)	3.3	3.6	5.5	10	7.6	
U3 - Toxicity prior Beta(1, 1)	3.4	3.7	5.5	10.5	6.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.8	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.7	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.8	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.7	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.5	2.7	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.7	1.2	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.5	2.6	1.1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.6	1.5	2.6	1.1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	17	15.5	42.2	20.6	4.5	0.2
U3 - Toxicity prior Beta(0.15, 0.85)	21.1	16.1	41.9	17.6	3.1	0.2
U2 - Toxicity prior Beta(0.3, 0.7)	15.7	16	42.5	20.5	5.2	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	21.9	16.3	38.2	20	3.5	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	15.7	14.8	41.4	21	6.9	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	21.6	17.1	38	18.2	4.9	0.2
U2 - Toxicity prior Beta(1, 1)	15.4	16.2	37.8	22.7	7.7	0.2
U3 - Toxicity prior Beta(1, 1)	21.6	16.3	39.1	17.4	5.4	0.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.3	6.7	9.2	6.3	2.5	
U3 - Toxicity prior Beta(0.15, 0.85)	5.9	6.7	9.1	6.2	2.1	
U2 - Toxicity prior Beta(0.3, 0.7)	4.9	6.3	9.3	6.6	2.9	
U3 - Toxicity prior Beta(0.3, 0.7)	5.5	6.4	9.1	6.5	2.5	
U2 - Toxicity prior Beta(0.5, 0.5)	4.7	6.1	9.1	6.8	3.3	
U3 - Toxicity prior Beta(0.5, 0.5)	5.2	6.2	9	6.6	2.9	
U2 - Toxicity prior Beta(1, 1)	4.6	5.9	8.9	7	3.6	
U3 - Toxicity prior Beta(1, 1)	5.1	6	8.8	6.9	3.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.5	2.1	2	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.5	2.1	1.9	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.4	2	2	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.4	2	1.9	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.9	2	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.3	1.9	1.9	0.6	
U2 - Toxicity prior Beta(1, 1)	0.5	3	1.8	1.9	0.6	
U3 - Toxicity prior Beta(1, 1)	0.5	3.1	1.8	1.8	0.5	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9.6	39.2	34.5	13.6	3.1	0
U3 - Toxicity prior Beta(0.15, 0.85)	11.2	45.5	32	9.8	1.5	0
U2 - Toxicity prior Beta(0.3, 0.7)	9.1	41.2	32.2	13.4	4.1	0
U3 - Toxicity prior Beta(0.3, 0.7)	12.6	45.2	28.5	12.1	1.6	0
U2 - Toxicity prior Beta(0.5, 0.5)	9.2	40.5	31.4	15	3.9	0
U3 - Toxicity prior Beta(0.5, 0.5)	12.6	44.1	30.1	10.8	2.4	0
U2 - Toxicity prior Beta(1, 1)	8.2	39.1	32.7	15.5	4.5	0
U3 - Toxicity prior Beta(1, 1)	11.5	44.7	28.2	13.5	2.1	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.4	11.3	9	4.2	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	12.6	8.6	3.4	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	4.4	11.2	8.9	4.2	1.3	
U3 - Toxicity prior Beta(0.3, 0.7)	4.5	12.3	8.6	3.7	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	4.2	11.1	8.9	4.4	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	4.4	12.1	8.6	3.8	1	
U2 - Toxicity prior Beta(1, 1)	4.2	10.9	8.9	4.5	1.5	
U3 - Toxicity prior Beta(1, 1)	4.3	12	8.5	4.2	1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3.1	2.1	1.2	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.2	2	1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	3	2	1.2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3.1	1.9	1.1	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	3	2	1.2	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	3	2	1	0.2	
U2 - Toxicity prior Beta(1, 1)	0.2	2.9	1.9	1.1	0.2	
U3 - Toxicity prior Beta(1, 1)	0.2	3	1.9	1	0.2	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	24.6	44.4	15.7	6.3	2.7	6.3
U3 - Toxicity prior Beta(0.15, 0.85)	29.6	43.3	14.1	5.2	1.4	6.4
U2 - Toxicity prior Beta(0.3, 0.7)	23.3	44.8	15.8	7.4	3.5	5.2
U3 - Toxicity prior Beta(0.3, 0.7)	28.6	43.4	14	6.2	2	5.8
U2 - Toxicity prior Beta(0.5, 0.5)	22.3	44.2	17.5	7.2	3.7	5.1
U3 - Toxicity prior Beta(0.5, 0.5)	26.4	45.1	14.2	6.5	2.6	5.2
U2 - Toxicity prior Beta(1, 1)	21.2	45.2	17.1	7.9	3.8	4.8
U3 - Toxicity prior Beta(1, 1)	25.7	44.7	14.5	7	2.4	5.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9	10.2	5.5	3.5	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	9.4	10.2	5.4	3.4	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	8.2	10.1	5.9	3.8	1.7	
U3 - Toxicity prior Beta(0.3, 0.7)	8.7	10.1	5.6	3.7	1.5	
U2 - Toxicity prior Beta(0.5, 0.5)	7.7	9.8	6.1	4.1	1.9	
U3 - Toxicity prior Beta(0.5, 0.5)	8.3	9.8	5.8	4	1.7	
U2 - Toxicity prior Beta(1, 1)	7.7	9.6	6	4.3	2	
U3 - Toxicity prior Beta(1, 1)	8.2	9.7	5.8	4.1	1.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	4.5	1.5	1.1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	4.4	1.5	1.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	4.3	1.4	1.1	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	4.3	1.4	1	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.8	1.2	1	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	3.8	1.2	1	0.2	
U2 - Toxicity prior Beta(1, 1)	0.7	3.3	1.2	0.9	0.2	
U3 - Toxicity prior Beta(1, 1)	0.7	3.3	1.2	0.9	0.2	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	25.8	52.1	15.5	3.6	0.3	2.7
U3 - Toxicity prior Beta(0.15, 0.85)	30.4	52	11.9	2.7	0.1	2.9
U2 - Toxicity prior Beta(0.3, 0.7)	25.7	52.2	15.2	3.8	0.7	2.4
U3 - Toxicity prior Beta(0.3, 0.7)	29.7	51.2	13.1	3.1	0.5	2.4
U2 - Toxicity prior Beta(0.5, 0.5)	25.3	53	14.9	4.2	0.7	1.9
U3 - Toxicity prior Beta(0.5, 0.5)	30	50.1	14.3	3.2	0.3	2.1
U2 - Toxicity prior Beta(1, 1)	23	52.5	16.1	4.7	0.9	2.8
U3 - Toxicity prior Beta(1, 1)	29.2	50.2	14.5	3.2	0.1	2.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.4	11.4	5.9	3.1	1	
U3 - Toxicity prior Beta(0.15, 0.85)	8.9	11.3	5.6	3	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	8	11.1	6.2	3.5	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	8.4	11.2	5.9	3.3	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	7.6	10.9	6.4	3.6	1.3	
U3 - Toxicity prior Beta(0.5, 0.5)	8.2	11	6.1	3.5	1.2	
U2 - Toxicity prior Beta(1, 1)	7.4	10.8	6.5	3.7	1.4	
U3 - Toxicity prior Beta(1, 1)	8	10.8	6.2	3.6	1.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	4.8	1.6	1	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	4.8	1.5	1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	1	4.6	1.5	1	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	4.6	1.5	1	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	4.2	1.3	1	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	4.1	1.3	0.9	0.2	
U2 - Toxicity prior Beta(1, 1)	0.7	3.7	1.2	0.9	0.1	
U3 - Toxicity prior Beta(1, 1)	0.9	3.7	1.2	0.8	0.1	